

BUSINESS INTELLIGENCE AND TOOLS

ASSIGNMENT – 1

Aim

To perform data preprocessing, Exploratory Data Analysis (EDA), Star Schema modeling, and OLAP operations such as Slicing, Dicing, Drill-Down, and Roll-Up on a sales dataset using Microsoft Power BI.

Objectives

1. To preprocess and clean the sales dataset using Power Query.
2. To perform Exploratory Data Analysis and understand data patterns.
3. To design a Star Schema using fact and dimension tables.
4. To establish relationships between the fact and dimension tables in Power BI.
5. To perform OLAP operations such as Slicing, Dicing, Drill-Down, and Roll-Up.
6. To create interactive Power BI visuals and dashboards for multidimensional sales analysis.

Tools Required

- Microsoft Power BI Desktop
- Microsoft Excel
- Sales Dataset (.xlsx)

Dataset Used

DS46_200_Rows_Dataset (1).xlsx

The uploaded dataset contains 200 rows and 18 columns. It contains order/date, geography, customer, industry, product, sales-channel, quantity, pricing, discount, cost, delivery and satisfaction information.

Important Power BI Calculations

The uploaded dataset does not contain a direct Sales or Profit column. Therefore, the following measures are derived from the available fields so that the analysis remains traceable to the actual dataset:

- Gross Sales = Quantity × UnitPrice
- Net Sales = Gross Sales × (1 – DiscountPct / 100)
- Total Cost = Quantity × CostPrice
- Profit = Net Sales – Total Cost

Suggested DAX measures in Power BI:

```
Gross Sales = SUMX(FactSales, FactSales[Quantity] * FactSales[UnitPrice])
Net Sales = SUMX(FactSales, FactSales[Quantity] * FactSales[UnitPrice] * (1 - FactSales[DiscountPct] / 100))
Total Cost = SUMX(FactSales, FactSales[Quantity] * FactSales[CostPrice])
Profit = [Net Sales] - [Total Cost]
```

Part A: Data Preprocessing

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Step 1: Import the Dataset

7. Open Power BI Desktop.
8. Select Home → Get Data → Excel.
9. Select DS46_200_Rows_Dataset (1).xlsx.
10. Select the required sheet and choose Transform Data to open Power Query. Fig. 1:

Connecting Power BI to the Sales Dataset

Step 2: Inspect the Data

11. Review all columns and records in Power Query Editor.
12. Verify data types: OrderDate as Date and numerical fields as numeric.
13. Identify categorical fields such as Region, ProductCategory and SalesChannel. 14. Check the preview for missing or inconsistent values.

Step 3: Data Cleaning

15. Check for null/blank values.
16. Check duplicate records using Remove Rows → Remove Duplicates where required. 17. Rename fields only where necessary.
18. Correct data types.
19. Review categorical values for consistency.
20. Create derived measures in the Power BI model rather than altering the raw values. Dataset check: 0 missing cells were found and 0 duplicate rows were found in the uploaded file.

Step 4: Create the Star Schema

In Power BI, the cleaned transaction table can be treated as FactSales. Separate dimension tables can be created for Date, Geography, Product and Sales Channel. The fact table stores transaction-level measures while the dimensions provide descriptive fields for analysis.

POWER BI — STAR SCHEMA MODEL

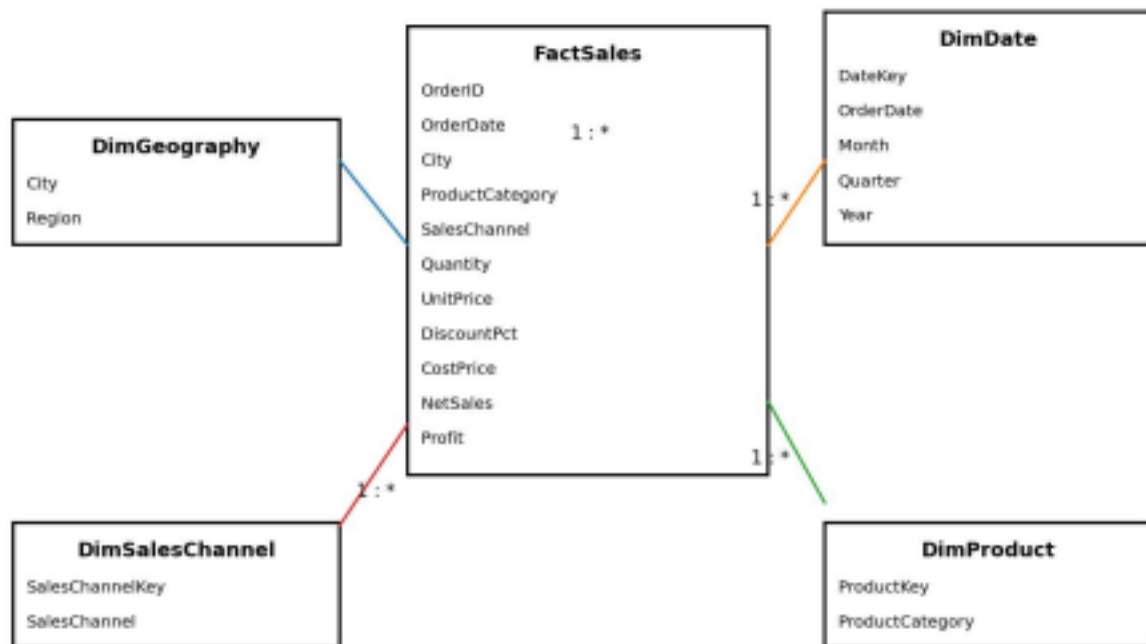


Fig. 2: Power BI Star Schema reference layout

Part B: Exploratory Data Analysis

Step 5: Data Profiling

21. Open Power Query and inspect column quality and distribution.

22. Check number of records, data types, missing values, unique values and distributions. 23. Use report visuals to examine business patterns.

Step 6: Analyze the Dataset

Important fields include Sales/Profit measures derived from Quantity, UnitPrice, DiscountPct and CostPrice, along with CustomerType, Industry, Region, ProductCategory, SalesChannel and OrderDate.

Step 7: Create Basic Visualizations

- Net Sales by Region
- Net Sales by Product Category
- Net Sales by Sales Channel
- Net Sales over time
- Profit analysis

POWER BI — SALES ANALYSIS DASHBOARD



Fig. 3: Power BI EDA dashboard reference

Part C: Star Schema Creation

Step 8: Create the Data Model

24. Open Model view in Power BI Desktop.
25. Add FactSales and the dimension tables.
26. Connect each dimension to FactSales using the relevant key/attribute.
27. Use one-to-many relationships with the dimension side on the '1' side and FactSales on the '*' side where the dimension values are unique.

Step 9: Establish Relationships

- FactSales[City] → DimGeography[City]
- FactSales[OrderDate] → DimDate[OrderDate]
- FactSales[ProductCategory] → DimProduct[ProductCategory]
- FactSales[SalesChannel] → DimSalesChannel[SalesChannel]

The exact relationship fields should be implemented only when the corresponding dimension columns contain unique values on the dimension side.

Part D: OLAP Operations

Step 10: Slicing

Slicing selects one value from a dimension to create a smaller subset of the dataset.

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28. Add Region to a slicer.

29. Select West.

30. Observe Net Sales, Profit and other visuals update for the selected region.

POWER BI — OLAP SLICING (Region = West)

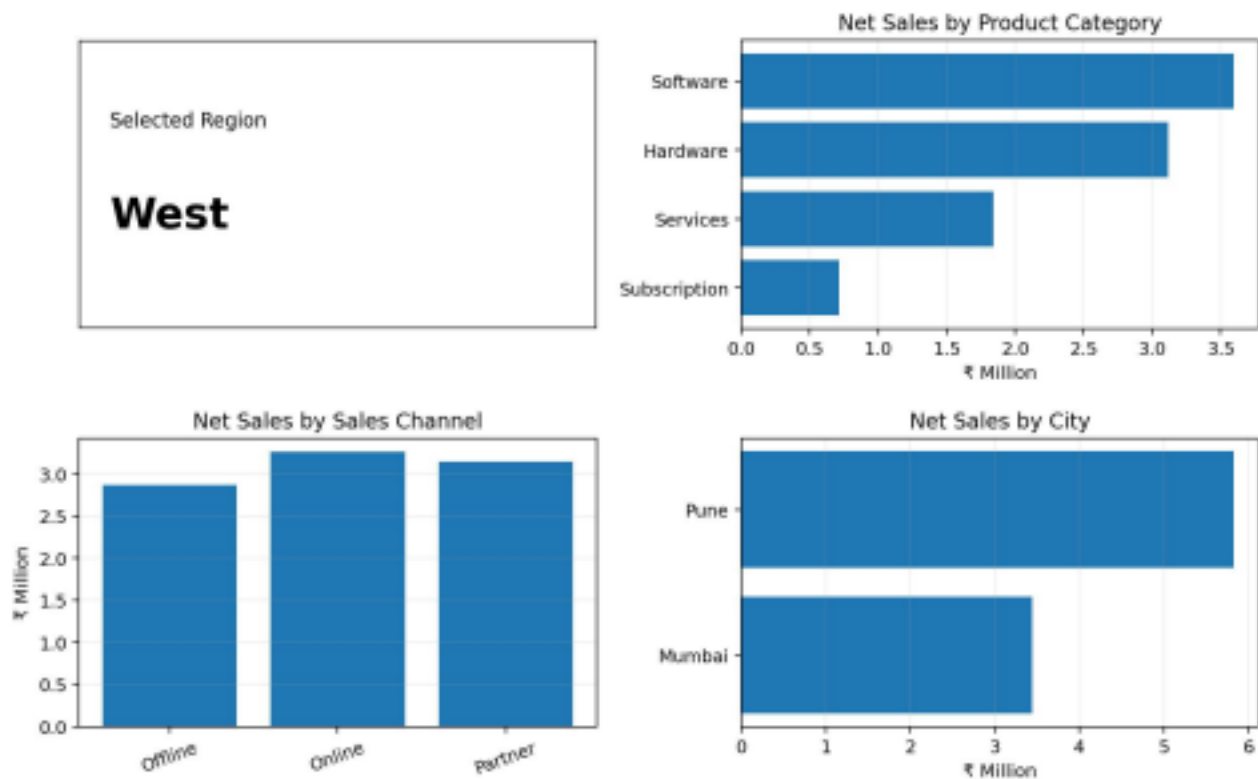


Fig. 4: Power BI OLAP Slicing reference

Step 11: Dicing

Dicing applies multiple dimension filters simultaneously to analyze a smaller multidimensional subset.

31. Add Region, ProductCategory and SalesChannel slicers.

32. Select West + Software + Online.

33. Observe the resulting Net Sales for the selected combination.

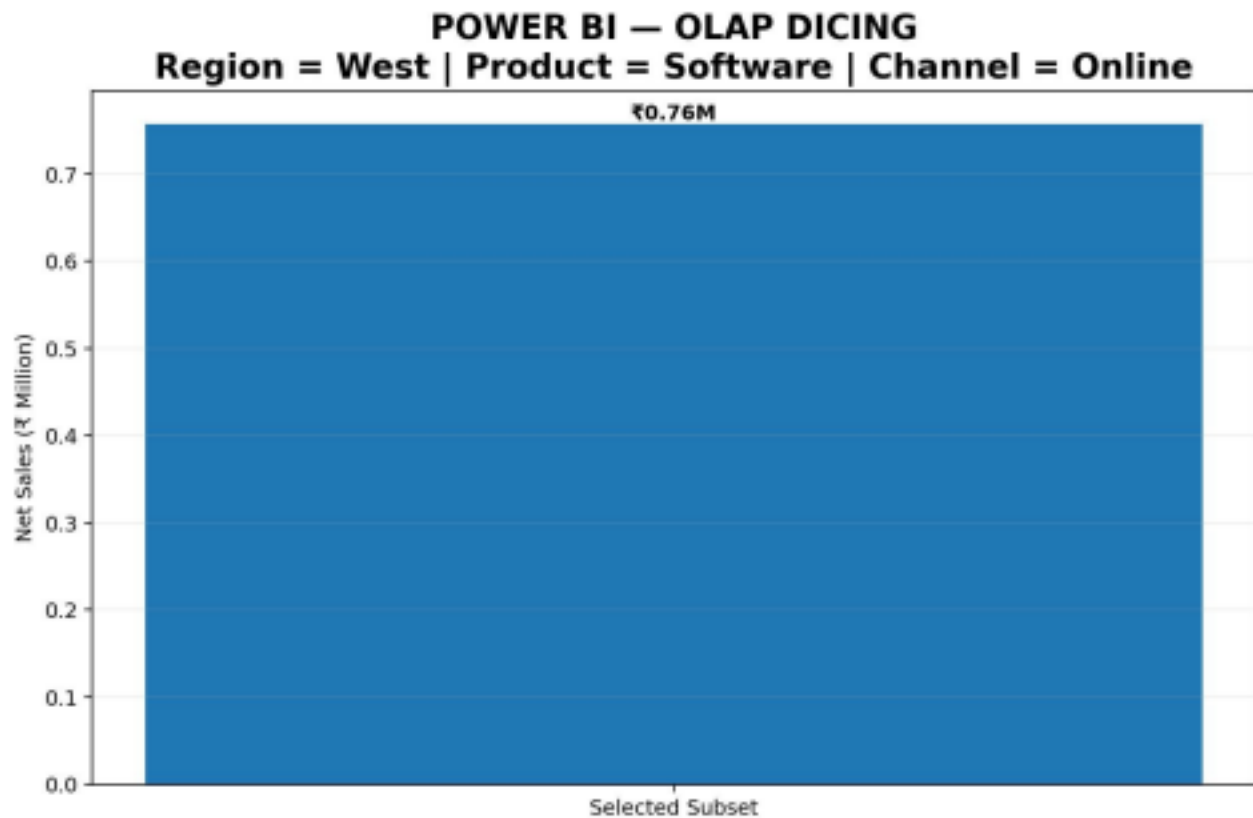


Fig. 5: Power BI OLAP Dicing reference

Step 12: Drill-Down

- 34. Create a Date table.
- 35. Create a hierarchy Year → Quarter → Month → Day.
- 36. Place the hierarchy in a visual.
- 37. Use the drill controls to move from Year to Quarter, Month and Day.
- 38. Observe increasingly detailed sales information.

Step 13: Roll-Up

- 39. Start from a detailed date level.
- 40. Use the drill controls to move from Day → Month → Quarter → Year.
- 41. Observe the aggregated values at each higher level.

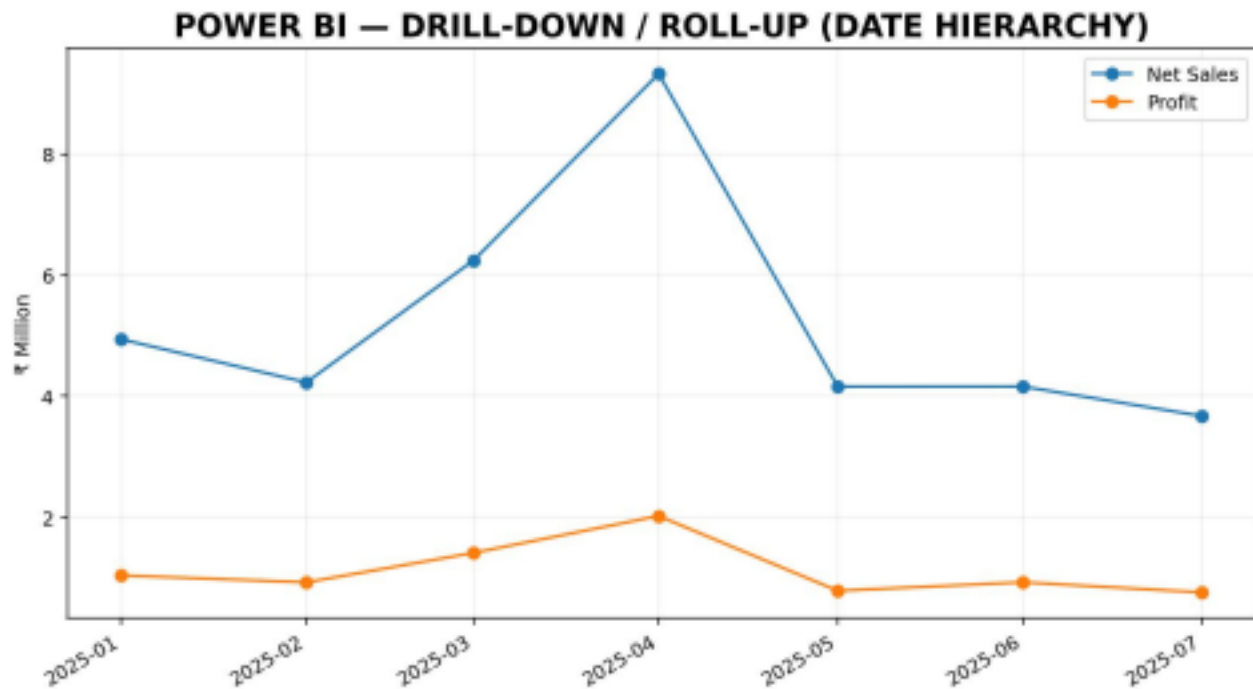


Fig. 6: Power BI Drill-Down / Roll-Up reference

Result

The sales dataset was successfully prepared for analysis in Power BI. Data preprocessing and profiling were performed, derived Net Sales and Profit measures were defined from the available fields, and a Star Schema structure was mapped using fact and dimension tables. OLAP operations including Slicing, Dicing, Drill-Down and Roll-Up were represented using Power BI slicers, filters, relationships, visuals and date hierarchies.

Learning Outcomes

1. I learned to perform data preprocessing and data cleaning using Power Query.
2. I understood the concepts of Exploratory Data Analysis and data profiling.
3. I learned to create Power BI visualizations for analyzing sales and profit patterns.
4. I gained practical knowledge of Star Schema modeling and relationships in Power BI.
5. I learned to implement Slicing and Dicing using Power BI slicers and filters.
6. I learned to perform Drill-Down and Roll-Up using date hierarchies.
7. I gained practical experience in designing interactive Power BI dashboards.
8. I understood the importance of OLAP operations in Business Intelligence and multidimensional analysis.